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FEROM: Frontier Endogenous Reveal-Order Marginal Policy Optimization for Masked Diffusion LMs

Marginalizing Over Reveal Orders Via a DAG Formulation Reduces Bias and Variance in Diffusion-LM Reinforcement Learning

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Masked diffusion language models iteratively unmask token positions, but applying reinforcement learning to them requires a policy gradient over exponentially many possible reveal orders. Fixed schedules introduce bias; single-trajectory sampling introduces high variance. FEROM Rao-Blackwellizes over all reveal orders using edge occupancy on a reveal-state DAG, cutting both failure modes through its Frontier Reveal Marginalization approximation.

AT A GLANCE

Problem: Policy gradient for masked diffusion LMs is either biased (fixed unmasking schedule) or high-variance (single sampled reveal order).

Why it matters: RLHF for masked dLLMs requires low-variance gradients; schedule bias and trajectory variance both degrade fine-tuning quality.

Method: Rao-Blackwellize over reveal orders via posterior edge occupancy on a reveal-state DAG; approximate tractably with Frontier Reveal Marginalization; integrate into a GRPO-style objective.

Result: Lower gradient variance, faster convergence, and higher reward than fixed-schedule and single-path GRPO baselines.

Evidence: Masked dLLM RL fine-tuning experiments with reward-guided generation tasks; comparisons against fixed-schedule and single-path GRPO.

The Big Picture

from a fully masked sequence and iteratively choosing which positions to reveal. At each step the model assigns a confidence to each masked position and reveals the most confident ones. Training with RL requires differentiating through this process to maximize an external reward.

The core difficulty is that the reveal order—which positions are unmasked in which step—is a discrete latent variable with a space that grows factorially in sequence length. Standard RL methods handle this either by fixing the unmasking order in advance (introducing bias, since the fixed schedule may not match the optimal order) or by sampling a single reveal order per rollout (introducing high variance, since any single path captures a tiny fraction of the reveal-order space).

FEROM reframes this as a Rao-Blackwellization problem: instead of evaluating the policy gradient on a single sampled path, marginalize over *all* valid reveal orders. The key insight is that this marginalization can be expressed analytically using the structure of the reveal-state DAG, and approximated tractably by restricting attention to the "frontier"—the boundary between revealed and masked positions.

Why Existing Approaches Fall Short

Fixed-schedule policy gradient assigns an unmasking order at the start of each rollout (e.g., lowest-confidence first, or uniform random). The gradient is computed along this fixed path. The bias arises because the fixed schedule does not adapt to the learned policy; the optimal reveal order changes as the model improves.

Single-path GRPO samples one reveal order per rollout. The variance of this estimator is proportional to the variance of the policy's log-probability over the reveal-order space, which is large when the model assigns similar probability to many orders.

Beam search over reveal orders approximates the best order greedily but is not differentiable and cannot be used for gradient estimation. FEROM is the first method to Rao-Blackwellize over reveal orders, strictly reducing variance relative to single-path sampling with no additional bias.

Core Mechanism

directed acyclic graph (DAG) over masking states. Each node $m \in \{0,1\}^T$ is an indicator vector of which positions are currently masked. An edge $(m \to m')$ exists when m' is obtained from m by unmasking exactly one additional position. A full reveal order is a path from the all-masked node to the all-revealed node.

For a rollout ending at a final answer y, FEROM replaces the single-path estimator with the Rao-Blackwellized estimate:

$$\nabla_\theta \log p_\theta(y) = \sum_{(m \to m') \in \mathrm{DAG}} \rho(m \to m' | y)\, \nabla_\theta \log p_\theta(m' | m, x)$$

where $\rho(m \to m' | y)$ is the *posterior edge occupancy*: the probability under the model that the transition $m \to m'$ occurred, given that the final output is y. This sum has the same expectation as the single-path estimator but strictly lower variance by Rao-Blackwell's theorem.

Technical Formulation

The posterior edge occupancy for edge $(m \to m')$ where $m' = m \setminus \{j\}$:

$$\rho(m \to m' | y) = \frac{\left[\sum_{\sigma:\ (m \to m') \in \sigma} q_\theta(\sigma | x)\, p_\theta(y | \sigma, x)\right]}{\left[\sum_{\sigma} q_\theta(\sigma | x)\, p_\theta(y | \sigma, x)\right]}$$

Exact computation is intractable. **Frontier Reveal Marginalization (FRM)** approximates it by restricting to the frontier $\mathcal{F}(m)$—the set of currently masked positions—and treating the marginal at each frontier position independently. The GRPO-style objective is:

$$\mathcal{L}_{\mathrm{FEROM}} = -\mathbb{E}_y[r(y)] \cdot \sum_{(m \to m')} \tilde{\rho}(m \to m' | y) \log q_\theta(j | \mathcal{F}(m), x)$$

Learning or Inference Procedure

FEROM is a training-time modification; at inference, the model samples normally. During training:

1. Sample rollouts $\{y_i\}$ from the current model using the standard masked-diffusion sampling procedure.
2. For each rollout, compute the FRM frontier occupancies $\tilde{\rho}(m \to m' | y_i)$ for each unmasking step.
3. Compute the FEROM gradient using the edge-based surrogate.
4. Apply the gradient update with a standard optimizer (Adam).

No additional model is required; FRM uses only the model's own logits at each frontier state.

What the Guarantee Says

is that for any rollout budget, the FEROM gradient estimator has strictly lower variance than the single-path GRPO estimator (by Rao-Blackwell), and is unbiased relative to the true policy gradient under the fixed-budget FRM approximation. The bias introduced by FRM's frontier approximation is not formally bounded but is empirically shown to be small.

Experimental Findings

- **Gradient variance:** FEROM reduces policy gradient variance by 40–65% compared to single-path GRPO across tasks and model sizes.
- **Convergence speed:** FEROM reaches target reward in 30–50% fewer gradient steps than fixed-schedule and single-path baselines.
- **Final reward:** At equal step budgets, FEROM achieves higher final reward on all evaluated reward-guided generation tasks.
- **FRM budget sensitivity:** Using $|\mathcal{F}| = 4$ frontier positions captures >90% of the variance reduction from exact marginalization; larger frontiers add diminishing returns.

Ablations and Interpretation

FRM vs. exact marginalization: On short sequences (length ≤ 16) where exact marginalization is tractable, FRM achieves 94% of the exact variance reduction, confirming that the frontier approximation is tight.

Frontier width: Increasing frontier width from 1 to 4 reduces variance by an additional 32%; width 8 adds only 4% more. Width 4 is used in all reported experiments.

Reward model quality: FEROM's advantage over single-path GRPO grows with reward noise: when the reward model is noisy, the lower gradient variance of FEROM prevents the optimizer from chasing reward-model artifacts.

Comparison to fixed schedule: Fixed reveal order eliminates reveal-order variance but introduces systematic bias that widens as training progresses (as the optimal policy deviates from the fixed schedule's implicit preference). FEROM's bias remains constant and near zero throughout training.

Reference

NeurIPS 2026 Submission. **FEROM: Frontier Endogenous Reveal-Order Marginal Policy Optimization for Masked Diffusion LMs.** arXiv:2611.10369. NeurIPS 2026. arxiv.org/abs/2611.10369